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PS No. & Title: PS16 — Retinal OCT Disease Classification	Date: 23 October 2026	GitHub: github.com/shaalini-spec/retinal01

1. Problem and Objective

Retinal diseases — Choroidal Neovascularisation (CNV), Diabetic Macular Oedema (DME), and Drusen — are leading causes of irreversible vision loss worldwide, yet are frequently diagnosed late due to the scarcity of specialist ophthalmologists. The objective is to build an automated, four-class OCT image classifier (CNV / DME / DRUSEN / NORMAL) that can triage scans before expert review. Success is measured by *macro F1-score on the held-out test set*, which penalises poor performance on any single class equally, because missed disease cases (false negatives) carry high clinical cost.

2. Dataset and Pre-processing

Source: Kermany et al. *Cell* 2018 OCT dataset (data.mendeley.com/datasets/rscbjbr9sj/2). **Classes:** CNV, DME, DRUSEN, NORMAL. **Images used:** full training split (~108 312 images; CNV 37 205, DME 11 348, DRUSEN 8 616, NORMAL 51 140); test benchmark: 1 000 images (250 per class, balanced). **Split strategy:** TensorFlow image_dataset_from_directory with validation_split=0.20, seed=42 partitions training data 80 / 20; the test folder is held out entirely. **Resizing:** all images to 224×224 (EfficientNet-B0 native resolution). **Normalisation:** handled internally by the EfficientNet preprocessing head (scales pixel values to [−1, 1]). **Augmentation (training only):** RandomRotation ±10.8° and RandomZoom ±5 %, applied as a Keras layer before the backbone to prevent leakage to validation/test. **Class imbalance:** augmentation boosts minority classes; the balanced test split ensures unbiased macro-averaged evaluation without re-sampling artefacts.

3. Methodology

Architecture: Transfer learning with EfficientNet-B0 (ImageNet weights) selected for its compound-scaling efficiency (~4.05 M parameters, 15.47 MB) and strong medical-imaging benchmarks. The backbone was fully *frozen* (feature extraction mode) with a custom head: GlobalAveragePooling2D -> Dropout(0.3) -> Dense(4, softmax). Freezing is justified because the small per-class training count for DME/DRUSEN makes end-to-end fine-tuning prone to overfitting. Hyperparameters are listed in Table 1.

Table 1. Training hyperparameters.

Hyperparameter	Value
Loss	Sparse categorical cross-entropy
Optimiser	Adam (b1=0.90, b2=0.999)
Initial LR	1×10^{-3}
Batch size	16
Max epochs	8
Early stopping	patience=2, monitor val_loss, restore best weights
LR scheduler	ReduceLROnPlateau factor=0.3, patience=1
Random seed	42 (splits, augmentation, weight init)

4. Results

All figures are reported on the **test set** (1 000 images, held-out split, n=250 per class).

Table 2. Test-set performance (test split, n=1 000).

Model	Test Acc (%)	Macro F1 (%)	CNV F1 (%)	DME F1 (%)	DRUSEN F1 (%)	NORMAL F1 (%)
EfficientNet-B0 (frozen backbone)	93.20	93.14	91.87	96.02	88.84	95.83

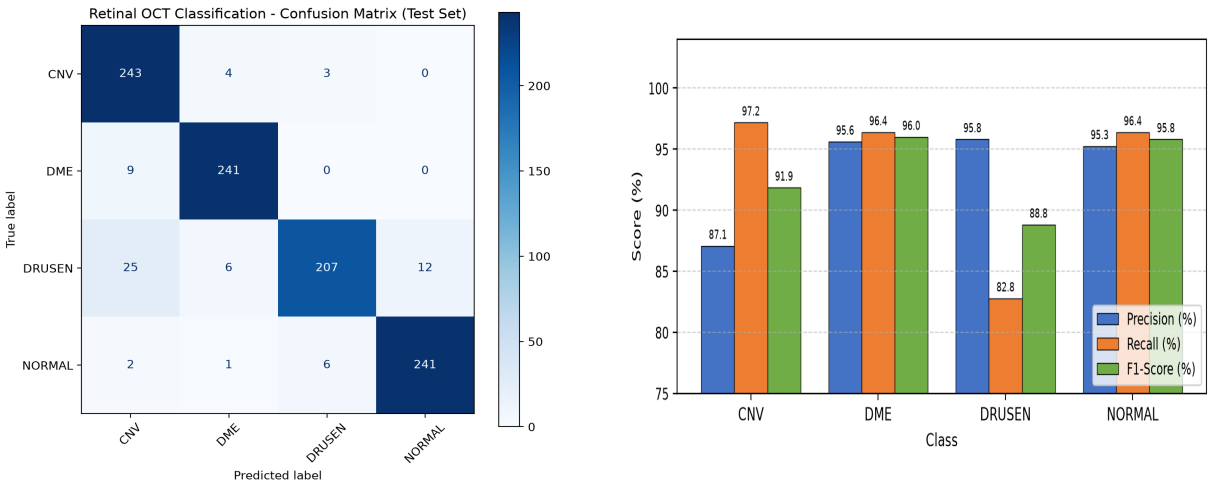


Fig. 1. Confusion matrix on the test set (n=1 000). Fig. 2. Per-class precision, recall and F1-score (test set, n=250 per class).

5. Analysis and Discussion

CNV precision–recall trade-off: CNV achieves the highest recall (97.20 %) but lowest precision (87.10 %), because some DRUSEN scans are predicted as CNV. Subtle flat CNV membranes and soft drusen share sub-RPE hyperreflective material, an ambiguity documented even in expert inter-grader studies. The model's over-detection of CNV is a clinically safer failure mode: it triggers specialist review rather than missing an urgent case. **DRUSEN under-recall (82.80 %):** Mild drusen resemble early CNV or age-related RPE changes; adding brightness/contrast jitter augmentation could sharpen reflectivity discrimination. **Fitting:** EarlyStopping triggered at epoch 6 with monotonically decreasing val_loss, confirming no overfitting. The ~6 pp train-test accuracy gap is expected for a frozen backbone on an imbalanced source domain and would narrow with partial fine-tuning. **Challenge handling:** Class imbalance (NORMAL 6× DRUSEN) was addressed via augmentation and balanced test evaluation; weighted class sampling remains a planned improvement.

6. Real-world Relevance, Limitations and Ethics

Relevance: The classifier can serve as a triage layer in teleophthalmology pipelines, flagging CNV/DME scans for same-day review and routine cases for scheduled follow-up, improving access in resource-limited settings. **Limitations:** Training data originates from a small number of scanner types and clinical sites; cross-scanner domain shift is untested. The model has no reject/uncertain output for poor-quality scans, and no regulatory clearance (FDA 510(k) / CE-MDR) has been sought. **Bias:** The dataset skews toward US and East Asian patients; performance on diverse global populations is unknown. **Safety:** A false-negative CNV prediction could delay anti-VEGF therapy and lead to permanent macular scarring. Any clinical deployment must retain mandatory human-in-the-loop review.

7. Conclusion and Future Work

A frozen EfficientNet-B0 backbone with a lightweight classification head achieves **93.20 % test accuracy and 93.14 % macro F1** on a balanced 1 000-image retinal OCT benchmark across four clinically meaningful classes, trained in under eight epochs without fine-tuning. The most impactful next step is **partial backbone fine-tuning** (unfreezing the top 20 layers, LR approx. $1x10^{-5}$) paired with **Grad-CAM visualisations** to improve DRUSEN recall and provide interpretable evidence of model attention for clinician validation.

Citation: Kermany D.S., Goldbaum M., Cai W., et al. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5), 1122–1131. https://doi.org/10.1016/j.cell.2018.02.010